

Rajalakshmi Engineering College

Name: Jeevan B
Email: 240701211@rajalakshmi.edu.in
Roll no: 2116240701211
Phone: 9025067651
Branch: REC
Department: CSE - Section 6
Batch: 2028
Degree: B.E - CSE

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 7_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

A financial analyst, Alex, needs a program to calculate simple interest for various financial transactions. He requires a straightforward tool that takes in the principal amount, interest rate, and time in years and computes the interest.

The formula to be used is: $\text{Interest} = \text{Principal} \times \text{Rate} \times \text{Time} / 100$

Implement this functionality using the InterestCalculator interface and the SimpleInterestCalculator class.

Input Format

The first line of input consists of the principal amount P as a double value.

The second line of input consists of the annual interest rate r as a double value.

The third line of input consists of the number of years t as a positive integer, which is an integer value.

Output Format

The output displays the calculated simple interest in the following format: "Simple Interest: [interest_value]", Here, [interest_value] should be replaced with the actual interest value calculated by the program.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 1000.00

5.00

2

Output: Simple Interest: 100.0

Answer

```
import java.util.Scanner;
```

```
// You are using Java
```

```
interface InterestCalculator{
```

```
    double simpleInterest(double principal,double rate,int time);
```

```
}
```

```
class SimpleInterestCalculator implements InterestCalculator{
```

```
    public double simpleInterest(double prin,double rate,int time){
```

```
        return (prin*rate*time/100);
```

```
    }
```

```
}
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        double principal = scanner.nextDouble();
```

```
        double rate = scanner.nextDouble();
```

```
        int time = scanner.nextInt();
```

```
InterestCalculator calculator = new SimpleInterestCalculator();  
double interest = calculator.simpleInterest(principal, rate, time);  
System.out.println("Simple Interest: " + interest);  
}  
}
```

Status : Correct

Marks : 10/10